

10(a)	<table border="1" data-bbox="353 213 1877 379"> <tr> <th data-bbox="353 213 1032 277">motor</th><th data-bbox="1032 213 1122 277"></th><th data-bbox="1122 213 1877 277">sensory</th></tr> <tr> <td data-bbox="353 277 1032 379">1 transmits impulses from CNS to effector</td><td data-bbox="1032 277 1122 379">and</td><td data-bbox="1122 277 1877 379">transmits impulses from, receptors / (named) sense organs, to CNS</td></tr> </table> <i>plus two from:</i> <table border="1" data-bbox="353 448 1877 842"> <tr> <th data-bbox="353 448 1032 512">motor</th><th data-bbox="1032 448 1122 512"></th><th data-bbox="1122 448 1877 512">sensory</th></tr> <tr> <td data-bbox="353 512 1032 608">2 cell body at end of, neurone / axon</td><td data-bbox="1032 512 1122 608">and</td><td data-bbox="1122 512 1877 608">cell body, in middle of / part-way along, neurone OR cell body not at end of, neurone / axon</td></tr> <tr> <td data-bbox="353 608 1032 671">3 cell body in, CNS / brain / spinal cord</td><td data-bbox="1032 608 1122 671">and</td><td data-bbox="1122 608 1877 671">cell body in (dorsal root) ganglion</td></tr> <tr> <td data-bbox="353 671 1032 767">4 long axon OR axon</td><td data-bbox="1032 671 1122 767">and</td><td data-bbox="1122 671 1877 767">short axon OR axon and dendron</td></tr> <tr> <td data-bbox="353 767 1032 842">5 dendrites attached to cell body</td><td data-bbox="1032 767 1122 842">and</td><td data-bbox="1122 767 1877 842">dendrites attached to dendron</td></tr> </table>	motor		sensory	1 transmits impulses from CNS to effector	and	transmits impulses from, receptors / (named) sense organs, to CNS	motor		sensory	2 cell body at end of, neurone / axon	and	cell body, in middle of / part-way along, neurone OR cell body not at end of, neurone / axon	3 cell body in, CNS / brain / spinal cord	and	cell body in (dorsal root) ganglion	4 long axon OR axon	and	short axon OR axon and dendron	5 dendrites attached to cell body	and	dendrites attached to dendron	3
motor		sensory																					
1 transmits impulses from CNS to effector	and	transmits impulses from, receptors / (named) sense organs, to CNS																					
motor		sensory																					
2 cell body at end of, neurone / axon	and	cell body, in middle of / part-way along, neurone OR cell body not at end of, neurone / axon																					
3 cell body in, CNS / brain / spinal cord	and	cell body in (dorsal root) ganglion																					
4 long axon OR axon	and	short axon OR axon and dendron																					
5 dendrites attached to cell body	and	dendrites attached to dendron																					
10(b)	<i>any three from:</i> 1 no myelin slow and myelin fast(er) / myelin increases speed ; 2 no myelin = $0.5\text{--}2\text{ m s}^{-1}$ and with myelin > 5 / up to 120 m s^{-1} ; 3 as axon diameter increases speed increases ; 4 diameter (+ units once) and speed (+ units once) for two neurones ;	3																					

Question	Answer	Marks
10(c)	any three from: A – any one reason from: 1 Na⁺ / sodium, channels already open OR Na⁺ has already entered neurone OR no more Na⁺ channels to open OR less Na⁺ outside to diffuse in OR less steep Na⁺ concentration gradient ; B – any one reason from: 2 sodium channels are, inactive / unresponsive OR potassium channels are open OR membrane is, impermeable / less permeable, to Na⁺ OR membrane is more permeable to K⁺ ; C – any one reason from: 3 harder to reach threshold OR potassium channels are (still) open OR sodium-potassium pumps need to restore the resting potential ; 4 hyperpolarisation at C ;	3